Phsysics4LLM reproducibility study

Syntetic data & Canon layer ablations

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The Architecture Zoo Problem

Transformer variants keep multiplying:

- Attention: sliding-window, MLA (DeepSeek), gated attention, DSA
- Positional encoding: RoPE, NoPE, ALiBi
- Gating: gated MLP, SwiGLU, MoE

Why can't we just compare them?

- Each change requires full re-training from scratch
- Academic scale (1.3B / 100B tokens): seed variance $\geq$ 2–4% swamps architecture differences of 1–2%
- Result: architecture ablations are slow, noisy, and expensive

The Synthetic Playground

Idea: Decompose "intelligence" into atomic skills; design one controlled task per skill.

Advantages:

- Infinite high-quality data, zero cost
- Controlled difficulty — no skill mixing
- Short context, no CoT leakage
- Cheap enough to run at small scale

Design criterion: mental reasoning only, no CoT, short contexts, real-world relevant.

Physics 4.1 Task Suite

Depo	Reasoning depth
Brevo	Reasoning breadth
Capo	Knowledge capacity
Mano	Knowledge manipulation
Lano	Language structure

Setup: k-hop traversal over a directed permutation graph. No Chain-of-Thought allowed. Each node label can span multiple tokens.

Format:

$$\texttt{<bos>}\ x_1\ y_1 \quad x_2\ y_2 \quad \cdots \quad \texttt{<query\_}k\texttt{>}\ q \ \rightarrow\ a$$

Edges presented in *random* order. Model must chain k retrieval steps entirely in activations.

Why it's hard:

- max k determines the difficulty of the task
- No intermediate output tokens — all reasoning happens in the forward pass
- Must resolve depth-k pointer chains in a single pass

Task Brevo — Reasoning Breadth

Setup: Recursive DAG traversal — given root q, output all reachable nodes in topological order. Node labels can span multiple tokens.

- Parallel fan-out: multiple reasoning threads must be held simultaneously
- Model must compute the full topological sort *before* emitting the first output token
- Graph size: up to 110 nodes

Motivation: Standard attention uses the *full* global mechanism just to copy information to a neighbouring token.

Solution: Lightweight horizontal residual links across neighbouring tokens:

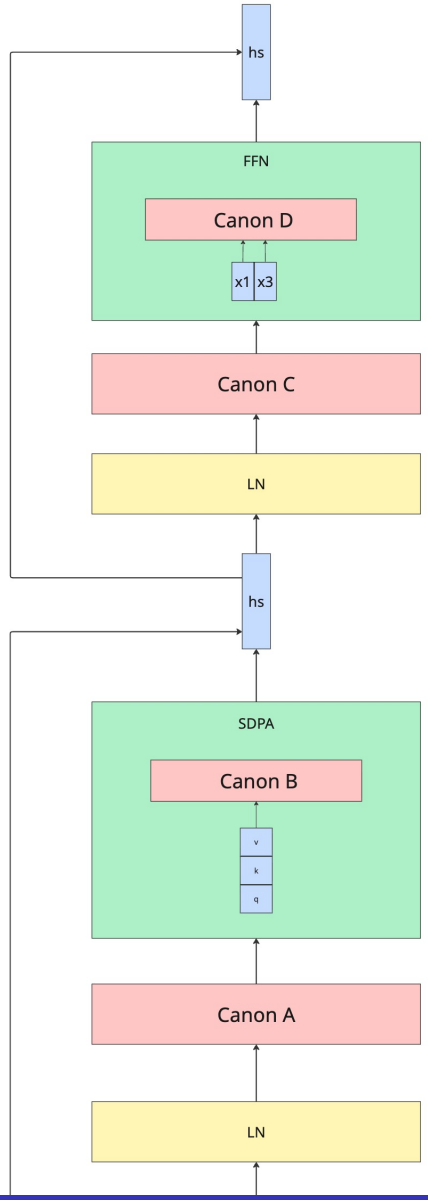
$$h'_t = h_t + w_0 \odot h_t + w_1 \odot h_{t-1} + w_2 \odot h_{t-2} + w_3 \odot h_{t-3}$$

Element-wise, trainable weights per layer.

Implementation:

- Causal `conv1d` with kernel size 4 + explicit residual
- $< 0.01\%$ extra parameters for a 1.3B model

Canon Layers — Insertion Points



4 insertion points per transformer block:

- **A** Pre-attention — before the attention sublayer
- **B** Q/K/V projections — inside attention, applied to query, key, and value
- **C** Pre-MLP — between attention output and the feed-forward sublayer
- **D** Inside MLP — between the two linear projections

Synthetic tasks (RoPE + Canon-ABCD):

- Reasoning depth: 2–4× improvement
- Reasoning breadth (Brevo): +30%
- Knowledge manipulation (Mano): +30%

Architecture comparisons:

- NoPE + Canon ≈ RoPE + Canon (length generalisation)
- GLA + Canon ≈ Mamba2 / GDN (simple baseline matches complex modern architectures)
- Real-world 1.3B / 100B tokens: Canon markedly improves NoPE and GLA

⇒ Synthetic task trends **mirror** real-world results.

Research Questions

- Can synthetic tasks be used to study training dynamics and the individual contribution of each architectural modification —
- Why does Canon help so much? Is it genuinely horizontal information mixing, or something else — like an implicit normalisation effect?
- How Canon parmeters impact performance?
  - **Initialization** — does the starting magnitude of the conv weights matter?
  - **Kernel size** — how many past tokens should Canon look at?
  - **Placement** — which layer depth and insertion point (A/B/C/D) drives the gain?

Reproducibility Setup

Framework: Meta Lingua (`lingua_modified`)

Model: Llama (RoPE), two sizes:

- 8 layers $\times$ 512 dim, 8 heads
- 12 layers $\times$ 768 dim, 12 heads

Tied input/output embeddings, `bf16`.

Training: 100 000 steps, AdamW, lr 3×10^{-4}, warmup 1 000 — single A100

Tasks & tokenization:

- Depo / Brevo: pass-through tokenizer, batch 256, seq len 768
- Text: bytes tokenizer, batch 128, seq len 2048

Loss: cross-entropy (only on answer tokens for synthetic tasks)

Findings

F1 **Training is noisy.** Seed variance dominates convergence speed; use final accuracy over multiple seeds.

F2 **Default init is not optimal.** $w_0 = 1/k$ beats zero, default random, and baseline; wrong init recovers slowly.

F3 **First canon layer captures most of the benefit.** Last-layer Canon $\approx$ baseline; first-layer Canon $\approx$ full Canon-ABCD.

F4 **Consistent improvement with along with any norm placement.** Pre-LN has the fastest convergence and biggest boost from canon.

F5 **Larger kernel size improves convergence.** ks $= 6 >$ 4 $>$ 2 $>$ baseline $>$ ks $= 1$; kernel of 1 is worse than no Canon.

F6 **Canon mixes information *and* normalises implicitly.** Higher weights on farther tokens; smoother gradients; fewer outlier activations.

Open Questions & Future work

- Training dynamics variance on synthetic data is determined by the task. What makes a task more noisy? How to design a task with training stability in mind?
- Is there a norm variation with training dynamics identical to canon.
- Reproduction with proper tokenizer.

Thank you!